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Duration	1 hour 59 mins
Grade	64.00 out of 85.00 (75.29%)

Question 1

Incorrect

Mark 0.00 out of 2.00

A hospital dataset contains the following patient record:

Patient ID	Age	Date of Birth	Blood Pressure
P102	42	03/07/2010	120

The data scientist must decide which data quality issue is MOST critical to address first before performing predictive modeling.

Select one:

- ☐ The record lacks completeness
- ☐ The record violates consistency
- ☐ The record lacks interpretability as date format is ambiguous
- ☒ The record violates believability because blood pressure is too normal. 
- ☐ The record lacks timeliness because blood pressure may change over time.

Your answer is incorrect.

The correct answer is: The record violates consistency

Question 2

Correct

Mark 2.00 out of 2.00

A bank is developing a classifier to predict loan default. The attribute Income has missing values. Customers belong to either enter the "High Risk" or "Low Risk" class.

Mean income values are:

- Overall mean income = \$45,000
- High Risk mean income = \$30,000
- Low Risk mean income = \$60,000

A High Risk customer has a missing income value.

Which replacement strategy is MOST likely to preserve class-specific information?

Select one:

- ☐ Replace with \$60,000 (Low Risk mean).
- ☐ Remove all High Risk customers with missing income.
- ☐ Replace with \$45,000 (global mean).
- ☐ Replace with "Unknown".
- ☒ Replace with \$30,000 (class-specific mean).  The lecture notes identify using the attribute mean for samples belonging to the same class as a smarter approach because it preserves class characteristics. Using the global mean may distort relationships between income and default risk.

Your answer is correct.

The correct answer is: Replace with \$30,000 (class-specific mean).

Question 3

Incorrect

Mark 0.00 out of 2.00

A company records monthly customer spending:

5, 8, 10, 12, 15, 18, 20, 24, 30, 200

The analyst applies equal-width binning with 5 bins.

Which statement is MOST accurate?

Select one:

- ☐ Equal-width binning guarantees every bin contains exactly two values.
- ☒ Equal-width and equal-depth binning will produce identical results. 
- ☐ Equal-width binning is preferred because it eliminates outliers automatically.
- ☐ In the Equal-width binning an outlier may greatly expand the data range.
- ☐ Equal-width binning will handle the extreme value 200 effectively because each bin contains similar numbers of records.

Your answer is incorrect.

The correct answer is: In the Equal-width binning an outlier may greatly expand the data range.

Question 4

Correct

Mark 2.00 out of 2.00

A financial institution has 150 attributes describing customers.

An investigation reveals:

- 30 attributes are exact duplicates of information already contained in other attributes.
- Analysts must preserve the original meaning of the remaining variables.

Which approach is MOST suitable?

Select one:

- ☐ PCA because it transforms all attributes into new dimensions
- ☐ Data Compression because fewer variables automatically remain
- ☒ Attribute Subset Selection because redundant attributes can be removed while preserving original variables. 
- ☐ Wavelet Transform because it preserves signal frequencies
- ☐ Factor Analysis because it identifies latent factors that explain correlations among attributes

Your answer is correct.

The correct answer is: Attribute Subset Selection because redundant attributes can be removed while preserving original variables.

Question 5

Correct

Mark 2.00 out of 2.00

A university is developing a model to predict student performance. The dataset contains the following records:

Student ID	GPA	Attendance (%)
S001	3.5	85
S002	Missing	82
S003	Missing	88
S004	Missing	90

Suppose 40% of the records in the dataset have missing GPA values, and GPA is an important predictor.

What is the MOST appropriate action?

Select one:

- ☐ Remove all records with missing GPA values.
- ☐ Ignore the missing values and proceed
- ☐ Replace all missing GPA values with 0.
- ☒ Estimate missing GPAs using inference-based methods such as decision trees or Bayesian approaches. ✔ Removing records would discard a large portion of the data and may introduce bias. The lecture notes identify inference-based methods as sophisticated approaches for estimating missing values while preserving underlying patterns.
- ☐ Replace all missing GPAs with the maximum observed GPA.

Your answer is correct.

The correct answer is: Estimate missing GPAs using inference-based methods such as decision trees or Bayesian approaches.

Question 6

Partially correct

Mark 8.00 out of 10.00

A retail company has collected customer data from multiple stores over several years. Before developing predictive models and generating business reports, the data science team performs several preprocessing activities,

Based on this scenario, which ONEs of the following are the most important reasons why data preprocessing is necessary

Select all applicable options. **Please note that one mark will be deducted for each incorrect option selected.**

Select one or more:

- ☒ Handles Imbalanced Data ✓
- ☒ Makes Variables Comparable ✓
- ☐ Prevents future data collection errors
- ☐ Removes the need for statistical analysis
- ☐ Increases the number of records automatically
- ☐ Guarantees perfect predictions
- ☒ Removes Noise and Outliers ✓
- ☒ Improves Interpretation ✓
- ☐ Ensures management approval of results
- ☒ Increases Model Accuracy ✓
- ☒ Makes all variables equally important ✗
- ☒ Prepares Data for Machine Learning Algorithms ✓
- ☐ Reduces Processing Time
- ☐ Guarantees machine learning success
- ☒ Ensures Consistency and Reproducibility ✓
- ☒ Handles Missing Values ✓
- ☐ Eliminates the need for domain knowledge
- ☐ Makes datasets look more professional
- ☒ Improves Data Quality ✓
- ☐ Makes reports longer and more detailed

Your answer is partially correct.

You have correctly selected 9.

The correct answers are: Improves Data Quality, Increases Model Accuracy, Handles Missing Values, Removes Noise and Outliers, Makes Variables Comparable, Prepares Data for Machine Learning Algorithms, Handles Imbalanced Data, Reduces Processing Time, Improves Interpretation, Ensures Consistency and Reproducibility

Question 7

Partially correct

Mark 1.00 out of 5.00

A university is conducting a study to identify factors that affect student academic performance. During data collection, the researchers discovered that some data values are missing. Read each scenario carefully and fill in the blank with the most appropriate term.

Scenario

The dataset contains only 1% missing values randomly distributed across records, and the sample size is very large. The researchers decide to remove the affected records. This method is called:

Pairwise Deletion.

**Scenario**

Researchers noticed that older students were less likely to disclose their monthly income. The missingness of income depends on the student's age, which is available in the dataset. The type of missingness is:

MNAR

**Scenario**

The dataset contains 40% missing values in the parental income variable, which is considered an important predictor of academic success. The researchers want to retain as much information as possible while obtaining reliable estimates. The most appropriate technique is:

Mean Imputation

**Scenario**

While transferring student records from one system to another, a software error randomly removed some values from different attributes. The missing values occurred completely by chance and were unrelated to any student characteristic. The type of missingness is :

MAR

**Scenario**

Students with very low GPAs were reluctant to report their GPA because they felt embarrassed about their performance. The probability of a GPA value being missing depends on the GPA itself. The type of missingness is:

MNAR



Your answer is partially correct.

You have correctly selected 1.

The correct answer is:

Scenario

The dataset contains only 1% missing values randomly distributed across records, and the sample size is very large. The researchers decide to remove the affected records. This method is called: → Listwise deletion,

Scenario

Researchers noticed that older students were less likely to disclose their monthly income. The missingness of income depends on the student's age, which is available in the dataset.

The type of missingness is: → MAR,

Scenario

The dataset contains 40% missing values in the parental income variable, which is considered an important predictor of academic success. The researchers want to retain as much information as possible while obtaining reliable estimates. The most appropriate technique is: → Multiple Imputation,

Scenario

While transferring student records from one system to another, a software error randomly removed some values from different attributes. The missing values occurred completely by chance and were unrelated to any student characteristic.

The type of missingness is : → MCAR,

Scenario

Students with very low GPAs were reluctant to report their GPA because they felt embarrassed about their performance. The probability of a GPA value being missing depends on the GPA itself.

The type of missingness is: → MNAR

Question 8

Partially correct

Mark 7.00 out of 10.00

nominal Binary indicator as.factor caret Labels factor levels categorical encoding machine learning
 Character predict() one-hot encoding ordered factor artificial ordering factor() ordinal presence
 reproducible pipeline dummyVars()

Enterprise Customer Analytics and Machine Learning Scenario

A large insurance company is developing a customer-retention prediction model. The data science team has received customer data from multiple branches. The dataset contains the following variables:

- Gender (Male, Female)
- Membership Type (Gold, Silver, Platinum)
- Province (Gauteng, Limpopo, Mpumalanga, etc.)
- Customer Satisfaction (Poor, Average, Good, Excellent)

The team discovers that several machine learning algorithms cannot process these variables directly because they are stored as text labels rather than numerical representations. The lead data scientist decides to apply appropriate categorical encoding techniques while ensuring that ordinal information is preserved where necessary and that artificial ordering is not introduced where it does not exist.

Complete the statements below by selecting the most appropriate answer from the list provided. Each correct answer is worth 1 mark.

Fill in the Blanks

1. Converting text categories into numerical representations before modelling is known as

categorical encoding ✓

2. Variables such as Province and Species are generally classified as nominal ✓ variables.

3. Variables such as Poor, Average, Good, and Excellent are ordinal ✓ variables.

4. The R function most commonly used to convert character variables into factors is factor() ✓

5. The categories stored within a factor object are known as factor levels ✓

6. When category order must be preserved explicitly, the analyst should create an ordered factor ✓ .

7. A major limitation of factor encoding is that some algorithms may incorrectly assume factor levels ✗ [artificial ordering] among categories that are actually unordered.

8. Labels ✗ [one-hot encoding] converts each category into a separate column rather than assigning category codes.

9. In the caret package, the function used to create an encoding model for indicator variables is dummyVars() ✓

10. After a dummy model has been created, the function commonly used to generate the encoded dataset is. one-hot encodi ✗ [predict()]

Your answer is partially correct.

Question 9

Partially correct

Mark 8.00 out of 10.00

A multinational retail company has consolidated transaction records from the United States, United Kingdom, and South Africa. The imported transaction date column contains the following

2025-09-01

01-10-2025

11-15-2025

2025-12-01

A junior analyst applies a single parsing function to the entire column and discovers that some records become missing after conversion. The chief data scientist concludes that the problem stems from inconsistent date representations and improper preprocessing.

Complete the following statements.

- The missing values generated during conversion are represented in R as ✓
- The date value "2025-09-01" requires the parser ✓
- The date value "01-10-2025" requires the parser ✓
- The date value "11-15-2025" requires the parser ✓
- Applying a single parser to differently formatted dates causes date ✓ errors.
- Before extracting components, the date variable should first be converted into a ✓ object.
- The process of creating Year, Month and Weekday variables from a Date variable is called ✓ engineering.
- To determine whether conversion failures occurred, the analyst should evaluate ✗
 (is.na(TransactionDate)).
- If the analyst wishes to identify the busiest day of the week for sales, the most suitable function is ✗
 ().
- Failure to correctly parse dates may introduce ✓ conclusions into subsequent business analyses.

Your answer is partially correct.

You have correctly selected 8.

The correct answer is:

A multinational retail company has consolidated transaction records from the United States, United Kingdom, and South Africa. The imported transaction date column contains the following

2025-09-01

01-10-2025

11-15-2025

2025-12-01

A junior analyst applies a single parsing function to the entire column and discovers that some records become missing after conversion. The chief data scientist concludes that the problem stems from inconsistent date representations and improper preprocessing.

Complete the following statements.

- The missing values generated during conversion are represented in R as [NA]
- The date value "2025-09-01" requires the parser [ymd()]

3. The date value "01-10-2025" requires the parser `[dmy()]`
4. The date value "11-15-2025" requires the parser `[mdy()]`
5. Applying a single parser to differently formatted dates causes date [parsing] errors.
1. Before extracting components, the date variable should first be converted into a `[Date]`
6. object.
7. The process of creating Year, Month and Weekday variables from a Date variable is called [feature] engineering.
8. To determine whether conversion failures occurred, the analyst should evaluate `[sum] (is.na(TransactionDate))`.
9. If the analyst wishes to identify the busiest day of the week for sales, the most suitable function is `[wday]()`.
10. Failure to correctly parse dates may introduce [misleading] conclusions into subsequent business analyses.

Question 10

Partially correct

Mark 34.00 out of 40.00

```

ggplot(comparison,
aes(x = Status,
y = price,fill = Status
data.frame(
price = diamonds_treated$price,
Status
par(mfrow = c(1, 2))
colSums(is.na(diamonds)) / nrow(diamonds) * 100
library(naniar)
diamonds
head(diamonds)
-5611.375
950
diamonds
data("diamonds")
nrow(diamonds),
step_impute_median(all_numeric_predictors()))
prep(diamond_recipe)
ggplot(diamonds,aes(y = price)
) +geom_boxplot(fill = "lightblue") +
labs(title = "Boxplot of Diamond Prices",y = "Price"
)
step_dummy(all_nominal_predictors())
summary(diamonds
library(ggplot2)
step_normalize(all_numeric_predictors())
str(diamonds)
pmin(
pmax(diamonds$price, Lower_Bound),
Upper_Bound
data.frame(
price = diamonds$price,
Status
colSums(is.na(diamonds))
3540
dim(diamonds)
ncol(diamonds),
5324.25
4374.25
step_nzv(all_predictors())
11885.62

```

Mandatory Practical Requirement

Note: Although some questions use drag-and-drop, fill-in-the-blank, or auto-marked formats, you are still required to complete the practical tasks in **Quarto** or **R Markdown**.

No marks will be awarded for answers that are not supported by practical work. Your submitted report must include the relevant **R code, outputs, visualisations, and interpretations** as evidence of how the answer was obtained.

Only the **rendered/knitted document** (PDF, Word, or HTML) will be accepted for assessment.

Brilliant Gems Jewellery (Pty) Ltd

Brilliant Gems Jewellery (Pty) Ltd is a rapidly growing jewellery retailer that sells diamonds through both physical stores and online platforms. The company maintains a large database containing information about each diamond, including its **price, carat weight, cut, colour, clarity, depth, table size, and dimensions**.

Management plans to implement a machine learning system to support pricing decisions, inventory planning, and customer recommendations. Before predictive models can be developed, the company's data science team must ensure that the data are clean, consistent, and suitable for analysis.

As a newly appointed **Junior Data Scientist**, you have been tasked with preparing the diamonds dataset for future analytics projects. Your responsibilities include exploring the dataset, identifying missing values, detecting and treating outliers, creating visualisations, and developing a reproducible preprocessing workflow using the **recipes** package.

The Head of Data Analytics has provided you with a partially completed R program. Your task is to complete the missing sections by writing the required code, executing it, and interpreting the results where requested.

Assessment Instructions

1. Read each question carefully.
2. Some questions provide **partially completed R code**. In these questions, you must **copy the code into your Quarto (.qmd) or R Markdown (.Rmd) document and complete the missing sections**.
3. Some questions provide the **expected output**. In these questions, you must write the R code that would generate the given output.
1. Other questions provide the **R code**. In these questions, you must execute the code and report or interpret the resulting output.
2. Where requested, provide short explanations and interpretations that relate your findings to the jewellery company's business objectives.
3. All code must be included in your report and must run successfully without errors.

Submission Requirements

You must complete the assessment using either:

- **Quarto (.qmd)**, or
- **R Markdown (.Rmd)**

Your submission must include:

- All completed code
- All outputs produced by the code
- All required visualisations
- All interpretations and discussions

After completing the assessment, **render (Quarto) or knit (R Markdown) your document** and submit **only the final rendered document** in one of the following formats:

- PDF
- Microsoft Word (DOCX)
- HTML

Important

Complete the missing sections of the partially written R programs by copying and pasting the code into your Quarto or R Markdown document.

Ensure that all code chunks execute successfully.

Ensure that all outputs and visualisations appear in the final document.

Submit only the rendered/knitted document.

title: "Test2"

author: "Student Number"

date: "2026-09-11"

output: word_document

#####

Question 1

Data Preprocessing Using the diamonds Dataset

Use The Diamond Dataset embedded in R

#####

```{r setup, include=FALSE}

knitr::opts\_chunk\$set(echo = TRUE)

---

# Load Required Packages

```{r }

library(ggplot2) ✓ # for professional-quality visualizations

library(naniar) ✓ # for Missing-value analysis and visualization

library(step) ✗ **[library(recipes)]** # Building reproducible data preprocessing workflows

Load Dataset

```{r , echo=FALSE}

# Load Dataset

data("diamonds") ✓

# Preserve original dataset

diamonds\_original ← diamonds ✓

# Reproducibility

set.seed(123) ✓

---

#####

# PART A: DATASET EXPLORATION

#####

```{r}

```

# First six records
head(diamonds) ✓

# Last ten records
tail(diamonds, 10) ✓

# Dimensions
str(diamonds) ✗ [dim(diamonds)]

# Number of observations
cat("Number of Observations:", nrow(diamonds), "\n") ✓

# Number of variables
cat("Number of Variables:", ncol(diamonds), "\n") ✓

# Structure
str(diamonds) ✓

# Summary statistics
summary(diamonds) ✓

...

#####
# PART B: MISSING VALUES
#####

```{r}
Missing values per variable
Missing <- colSums(is.na(diamonds)) ✓

Percentage missing values per variable
pctMissing <- colSums(is.na(diamonds)) / nrow(diamonds) * 100 ✓

Missing-value visualization
vis_miss(diamonds) ✓

...

#####
PART C: OUTLIER DETECTION AND TREATMENT
#####

#####
Question 5
#####

```{r}
Determine values of
Q1 = 950 ✓
Q3 = 5324.25 ✓
IQR = 4374.25 ✓
Lower Bound = -5611.375 ✓
Upper Bound = 11885.62 ✓

...

#####
# Question 6
#####

```{r}
...

Number of Outliers = 3540 ✓

#####
Question 7

```

```
#####
```

```
```{r}
```

```
# Perform Winsorization
```

```
diamonds_treated <- diamonds
```

```
diamonds_treated$price <- pmin(pmax(diamonds$price, Lower_Bound), Upper_Bound)
pmax(diamonds$price, Lower_Bound),
Upper_Bound]
```

```
)
...
```

```
#####
```

```
# Question 8
```

```
#####
```

```
```{r}
```

```
Write code to produce the following boxplot in a row . Use par() function
```

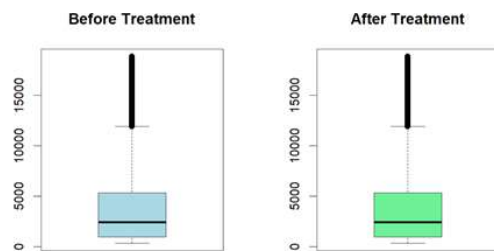
```
#par() function
```

```
par(mfrow = c(1, 2))
```

```
boxplot(diamonds$price,
main = "Before Treatment",
col = "lightblue"
)
```

```
boxplot(diamonds_treated$price,
main = "After Treatment",
col = "lightgreen"
)
```

```
...
```



```
#####
```

```
PART D: RECIPES AND DATA TRANSFORMATION
```

```
#####
```

```
#####
```

```
Question 9
```

```
#####
```

```
```{r}
```

```
# Complete the code below to create a preprocessing recipe that handles missing values, creates dummy variables,
normalizes numeric predictors, and removes near-zero variance predictors.
```

```
diamond_recipe <-
```

```
recipe(price ~ ., data = diamonds)
```

```
step_impute_median(all_numeric_predictors())
```

```
step_dummy(all_nominal_predictors())
```

```
step_normalize(all_numeric_predictors())
```

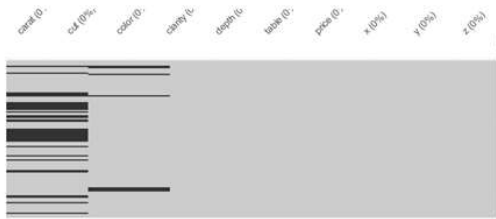
```
step_nzv(all_predictors())
```

```

diamond_recipe
# Prepare recipe
prep_recipe <- prep(diamond_recipe)
# Apply recipe
processed_data <- bake(
  prep_recipe,new_data = NULL
)
head(processed_data)
...

#####
# Question 11
#####
```{r}
#Visualize missing values
vis_miss(diamonds)
...

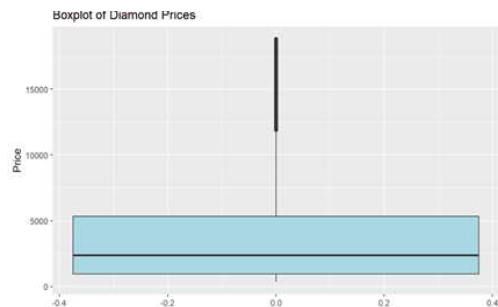
```



```

#####
Question 12
#####
```{r}
ggplot(diamonds,aes(y = price)) +geom_boxplot(fill = "lightblue") +labs(title = "Boxplot of Diamond Prices",y = "Price")
) +geom_boxplot(fill = "lightblue") +
labs(title = "Boxplot of Diamond Prices",y = "Price")
)
...

```



```

#####
# Question 13
#####
```{r}
before <-
price = diamonds$price,
>Status] = "Before"
)
after <-
price = diamonds_treated$price,

```

```
Status] = "After"
```

```
)
```

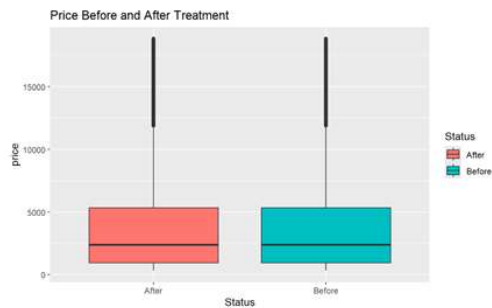
```
comparison <- [rbind(before, after)]
```

```
ggplot(comparison,aes(x = Status,y = price,fill = Status
aes(x = Status,
y = price,fill = Status]
```

```
)) +geom_boxplot() +labs(title = "Price Before and After Treatment"
```

```
)
```

```
...
```



```
#####
```

```
Question 14
```

```
#####
```

```
```{r}
```

```
# Demonstration of reproducibility
```

```
rbind(before,after) [set.seed(123)]
```

```
# Data preservation
```

```
diamonds_original <- diamonds
```

```
...
```

Your answer is partially correct.

Comment:

Remarked added 4

Question 11

Complete

Not graded

Upload File



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